Lumen Routing: Grounded Reading Companions for Long Papers

Enlumor Acceptance Fixture

Abstract. We introduce Lumen Routing, a method that first selects the passages supporting a claim and only then explains them, so every explanation can be traced back to its source sentence. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites.

1. Introduction

Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites.

2. Method

Lumen Routing has two stages. The router scores passages against the reader's question; the explainer writes an answer citing only the routed passages. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites.

3. Experiments

We compare three methods: Baseline, Retrieval and Lumen. As Figure 1 shows, one of them clearly leads. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites.

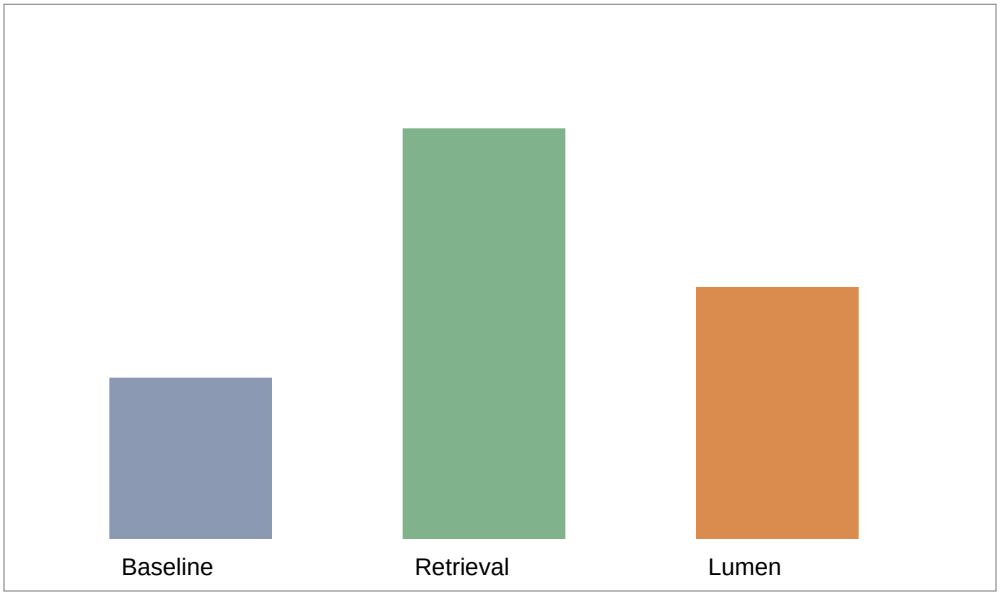


Figure 1: Accuracy of the three methods on the long-document benchmark.

4. Results

Table 1 lists the cost of each method.

Method	Tokens per answer	Latency (s)
Baseline	1,200	2.1
Retrieval	2,400	3.4
Lumen	1,900	2.8

Table 1: Cost per answer.

5. Conclusion

Grounding every explanation in routed passages keeps a reading companion honest. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites. Long academic documents spread one argument over many pages, so a reader must keep earlier claims in mind while checking new evidence. We study how a reading companion can stay grounded in the exact sentences it cites.

References

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